

L4: Update & Delete data

RECAP from Lab #3

- Insert Data

```
INSERT INTO Courses (  
course_id, course_name, credits)  
VALUES  
(101, 'Mathematics', 3),  
(102, 'Physics', 4),  
(103, 'Chemistry', 3);
```

```
INSERT INTO Students (student_id, name, dob, enrollment_date)  
VALUES (3, 'Emily Johnson Updated', '2001-08-26', NOW())  
ON DUPLICATE KEY UPDATE  
    name = VALUES(name),  
    enrollment_date = VALUES(enrollment_date);
```

```
INSERT INTO Students (  
student_id, name, dob, enrollment_date)  
SELECT  
student_id, name, dob, enrollment_date  
FROM TempStudents;
```

4.1 UPDATE statement

The UPDATE statement updates data in a table. It allows you to **change** the **values** in one or more columns of a single row or multiple rows.

Syntax

```
UPDATE [LOW_PRIORITY] [IGNORE] table_name
SET
    column_name1 = val1,
    column_name2 = val2,
    ...
[WHERE
    condition];
```

UPDATE statement (Modifier)

- The `LOW_PRIORITY` modifier instructs the `UPDATE` statement to **delay the update until there is no connection** reading data from the table. The `LOW_PRIORITY` takes effect for the storage engines that use table-level locking only such as `MyISAM`, `MERGE`, and `MEMORY`.
- The `IGNORE` modifier enables the `UPDATE` statement to **continue updating rows even if errors occurred**. The rows that cause errors such as duplicate-key conflicts are not updated.

UPDATE statement

Example

```
UPDATE
    employees
SET
    lastname = 'Hill',
    email = 'mary.hill@classicmodelcars.com'
WHERE
    employeeNumber = 1056;
```

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1056	Patterson	Mary	x4611	mpatterso@classicmodelc...	1	1002	VP Sales
	↓			↓			
employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1056	Hill	Mary	x4611	mary.hill@classicmodelcar...	1	1002	VP Sales

4.2 Replace String Function

Syntax

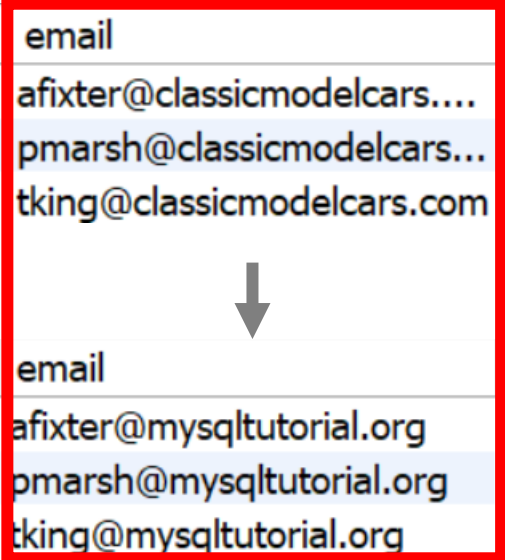
```
REPLACE(str_column,old_string,new_string);
```

Example

```
UPDATE
    employees
SET
    email =
        REPLACE(email,'@classicmodelcars.com','@mysqltutorial.org')
WHERE
    jobTitle = 'Sales Rep' AND    officeCode = 6;
```

Table: employees

employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1611	Fixter	Andy	x101	afixter@classicmodelcars....	6	1088	Sales Rep
1612	Marsh	Peter	x102	pmarsh@classicmodelcars...	6	1088	Sales Rep
1619	King	Tom	x103	tking@classicmodelcars.com	6	1088	Sales Rep



employeeNumber	lastName	firstName	extension	email	officeCode	reportsTo	jobTitle
1611	Fixter	Andy	x101	afixter@mysqltutorial.org	6	1088	Sales Rep
1612	Marsh	Peter	x102	pmarsh@mysqltutorial.org	6	1088	Sales Rep
1619	King	Tom	x103	tking@mysqltutorial.org	6	1088	Sales Rep

Note. You might get the error `SQL Error [1175] [HY000]: You are using safe update mode and you tried to update a table without a WHERE that uses a KEY column.` This could be in MySQL Workbench, or any other database GUI client. The error comes from the MySQL server, not the client.

Solutions:

```
SET SQL_SAFE_UPDATES = 0;
```

4.3 UPDATE JOIN statement

You can use the `JOIN` clauses in the `UPDATE` statement to update rows in one table based on values from another table.

Syntax

```
UPDATE
    T1
[INNER JOIN | LEFT JOIN] T2 ON T1.C1 = T2.C1
SET
    T1.C2 = T2.C2,
    T2.C3 = expr
WHERE
    condition;
```

UPDATE JOIN statement

Example

```
UPDATE
    employees
INNER JOIN merits ON employees.performance = merits.performance
SET
    salary = salary + salary * percentage;
```

Table: employees

emp_id	emp_name	performance	salary
1	Mary Doe	1	50000.00
2	Cindy Smith	3	65000.00
3	Sue Greenspan	4	75000.00
4	Grace Dell	5	125000.00
5	Nancy Johnson	3	85000.00
6	John Doe	2	45000.00
7	Lily Bush	3	55000.00

Table: merits

performance	percentage
1	0.00
2	0.01
3	0.03
4	0.05
5	0.08

Table: employees (New)

emp_id	emp_name	performance	salary
1	Mary Doe	1	50000.00
2	Cindy Smith	3	66950.00
3	Sue Greenspan	4	78750.00
4	Grace Dell	5	135000.00
5	Nancy Johnson	3	87550.00
6	John Doe	2	45450.00
7	Lily Bush	3	56650.00

4.4 DELETE JOIN statement

To delete rows from one table that has matching rows in another table.

Syntax

```
DELETE T1, T2  
FROM T1  
INNER JOIN T2 ON T1.key = T2.key  
WHERE condition;
```

DELETE JOIN statement (Matching)

You can use the INNER JOIN clause in the DELETE statement to delete rows from one table that **has matching rows** in another table.

Example

```
DELETE t2 FROM t1  
INNER JOIN t2  
ON t2.ref = t1.id ;
```

Table: T1

id
1
2
3

Table: T2

id	ref
A	1
B	2
C	3
D	4
E	5

Table: T2 (New)

id	ref
D	4
E	5

DELETE JOIN statement (Not matching)

We can also use the `LEFT JOIN` clause in the `DELETE` statement to delete rows in a table (left table) that **does not have matching rows** in another table (right table).

Example

```
DELETE T1
FROM T1 LEFT JOIN
      T2 ON T1.key = T2.key
WHERE
      T2.key IS NULL;
```

Example

```
DELETE customers
FROM customers
      LEFT JOIN
      orders ON customers.customerNumber = orders.customerNumber
WHERE
      orderNumber IS NULL;
```

Table: customers

customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1
361	Kommission Auto	Josephs	Karin	0251-555259	Luisenstr. 48
362	Gifts4AllAges.com	Yoshido	Juri	6175559555	8616 Spinnaker Dr.
363	Online Diecast Creations Co.	Young	Dorothy	6035558647	2304 Long Airport Avenue
369	Lisboa Souveniers, Inc	Rodriguez	Lino	(1) 354-2555	Jardim das rosas n. 32

Table: orders

orderNumber	orderDate	requiredDate	shippedDate	status	comments	customerNumber
10264	2004-06-30	2004-07-06	2004-07-01	Shipped	Customer will send a truck...	362
10295	2004-09-10	2004-09-17	2004-09-14	Shipped	They want to reevaluate t...	362
10414	2005-05-06	2005-05-13	NULL	On Hold	Customer credit limit exce...	362
10100	2003-01-06	2003-01-13	2003-01-10	Shipped	NULL	363
10192	2003-11-20	2003-11-29	2003-11-25	Shipped	NULL	363
10322	2004-11-04	2004-11-12	2004-11-10	Shipped	Customer has worked wit...	363

Example

```
DELETE customers
FROM customers
     LEFT JOIN
     orders ON customers.customerNumber = orders.customerNumber
WHERE
     orderNumber IS NULL;
```

Table: Query

customerNumber	customerName	orderNumber
361	Kommission Auto	NULL
362	Gifts4AllAges.com	10264
362	Gifts4AllAges.com	10295
362	Gifts4AllAges.com	10414
363	Online Diecast Creations Co.	10100
363	Online Diecast Creations Co.	10192
363	Online Diecast Creations Co.	10322
369	Lisboa Souveniers, Inc	NULL

Table: customers (New)

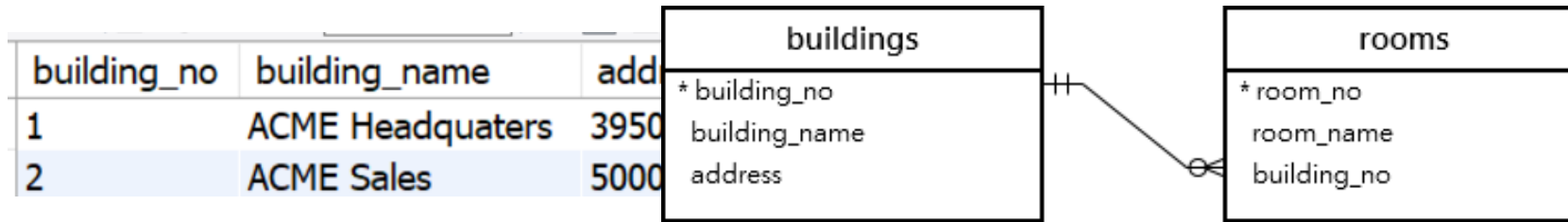
customerNumber	customerName	contactLastName	contactFirstName	phone	addressLine1
362	Gifts4AllAges.com	Yoshido	Juri	6175559555	8616 Spinnaker
363	Online Diecast Creations Co.	Young	Dorothy	6035558647	2304 Long Airp

4.5 ON DELETE CASCADE statement

This function allows you to **delete data from child tables automatically** when you delete the data from the parent table.

Example

```
CREATE TABLE rooms (  
    room_no INT PRIMARY KEY AUTO_INCREMENT,  
    room_name VARCHAR(255) NOT NULL,  
    building_no INT NOT NULL,  
    FOREIGN KEY (building_no)  
        REFERENCES buildings (building_no)  
        ON DELETE CASCADE  
);
```



room_no	room_name	building_no
1	Amazon	1
2	War Room	1
3	Office of CEO	1
4	Marketing	2
5	Showroom	2

DELETE FROM **buildings**
WHERE building_no = 2;

building_no	building_name	address
1	ACME Headquarters	3950 North 1st Street

room_no	room_name	building_no
1	Amazon	1
2	War Room	1
3	Office of CEO	1

[Tip] Find tables affected ON DELETE CASCADE

You can query this data from the `referential_constraints` in the `information_schema` database.

Example

```
USE information_schema;

SELECT
    table_name
FROM
    referential_constraints
WHERE
    constraint_schema = 'database_name'
    AND referenced_table_name = 'parent_table'
    AND delete_rule = 'CASCADE'
```

[Tip] Find tables affected ON DELETE CASCADE

Example

```
USE information_schema;

SELECT
    table_name
FROM
    referential_constraints
WHERE
    constraint_schema = 'lab5'
    AND referenced_table_name = 'buildings'
    AND delete_rule = 'CASCADE'
```

TABLE_NAME
rooms

4.6 TRUNCATE TABLE statement

Syntax

```
TRUNCATE [TABLE] table_name;
```

This function allows you to **delete all rows** from a table.

Same result as:

- 1) DELETE FROM table_name;
- 2) DROP TABLE table_name;
CREATE TABLE table_name (...);

room_no	room_name	building_no
1	Amazon	1
2	War Room	1
3	Office of CEO	1



room_no	room_name	building_no
NULL	NULL	NULL

If there is any FOREIGN KEY constraints from other tables that reference the table that you truncate, the TRUNCATE TABLE statement will fail.